

PRAMAAN

Explainable crime-scene evidence triage & FSL scheduling — an MCP server that IBM Bob drives

From a free-text seizure list to a ranked, flagged, sequenced and scheduled forensic-lab submission — with a tamper-evident chain of custody.

Team Pramaan

PRAMAAN

evidence triage · FSL scheduling · chain of custody

engine v1.0.0 · KB 858d6dd579 · LLM none · MCP /acp

+ New triage

Lab queueall cases

FSL networkscale-out

Benchmark200 exhibits

MCP servertools

CASES (4)

Cr. No. 784/2026, PS Shams...
P2 · 8 · homicide · B4EC8

Cr. No. 877/2026, PS Gachib...
P2 · 3 · hit and run · 29721

Cr. No. 1093/2026, PS Kukat...
P1 · 3 · burglary · 21561

Cr. No. 512/2026, PS Madha...
P1 · 5 · sexual assault · 82EA4

HOMICIDE / SUSPICIOUS DEATH · PRM-20260928-B4EC8

Cr. No. 784/2026, PS Shamshabad Rural (synthetic)

Roadside homicide with burning — culvert under service road

knowledge base 858d6dd5797d · engine v1.0.0 · LLM none
result sha-256 1c45f0eb0e2ff6ec ✓ matches ledger

EXHIBITS
18

P1 CRITICAL
8

P2 HIGH
8

HELD · STAGE 2
1

URGENT FLAGS
8

CUSTODY DEADLINE
25 Dec 2026

EVIDENTIAL VALUE RETAINED · P1 DAYS
96.1% vs 96.0% status quo · 2.06 vs 2.38 d

Act now — 6 field / preservation actions

Ex-A5 · PERISHABLE Degrades below usable quality in ~42 h under ambient storage. Move to an airtight container (unused paint can / nylon fire-debris bag) to extend this to ~104 days.

Ex-A17 · PERISHABLE Degrades below usable quality in ~42 h under ambient storage. Move to refrigeration (2-8 °C) to extend this to ~17 days.

Ex-A12 · COLLECT_NOW Still at the scene: evidence is likely lost in ~36 h (overwrite, weathering, washing). Collect immediately.

Ex-A13 · DIGITAL_OVERWRITE DVR/NVR storage loops and overwrites (often 7-30 days). Seize or export with hash today.

Ex-A11 · COLLECT_NOW Still at the scene: evidence is likely lost in ~36 h (overwrite, weathering, washing). Collect immediately.

Ex-A6 · PERISHABLE Degrades below usable quality in ~42 h under ambient storage. Move to an airtight container (unused paint can / nylon fire-debris bag) to extend this to ~104 days.

Evidence queue 38 · Lab plan 43 · Gaps 4 · Authenticity · BSA 3 · Custody ledger 7 · FSL packet

All · P1 · P2 · P3 · Held

■ probative ■ urgency ■ irreplaceable ■ statutory floor

#	EXHIBIT	DESCRIPTION	TYPE	TIER	EPI (0-100)	TIME TO LOSS	FLAGS
1	Ex-A8	bloodstained stone approx 2 kg near the body	Blunt object used as weapon	P1		61.8 22 d	
2	Ex-A2	half-burnt cigarette butt, 3 m from the body	Cigarette / beedi butt	P1		61.6 14 d	HEAT_DAMAGED_DNA
3	Ex-A5	empty petrol bottle (1 litre) found 5 m from the body	Accelerant container (petrol / kerosene)	P1		57.3 42 h	PERISHABLE · VOLATILE_TOXICANTS

Decisive evidence waits in a first-come, first-served queue

200+

exhibits seized in the 2019
Hyderabad case

3,000+

scene photographs to review

Weeks

typical FSL testing delay

60 / 90 days

custody clock, BNSS §187(3)

- A DNA-bearing cigarette butt that helped link the accused was nearly lost among roadside litter.
- BNSS 2023 §176(3) now mandates forensic scene visits for offences punishable with 7+ years — more scenes, same laboratories.
- Forwarding letters list exhibits in seizure order: no signal of what is decisive, perishable or deadline-bound.
- Officers decide what to seize, preserve and send within hours — without structured, defensible guidance.

Why evidence triage is hard

Needles in volume

Most exhibits are near-identical litter; a handful decide the case — and they look alike on a list.

Perishability varies 1000×

Wet swab: a day. Viscera at room temperature: two. CCTV: overwritten in days. A knife: years.

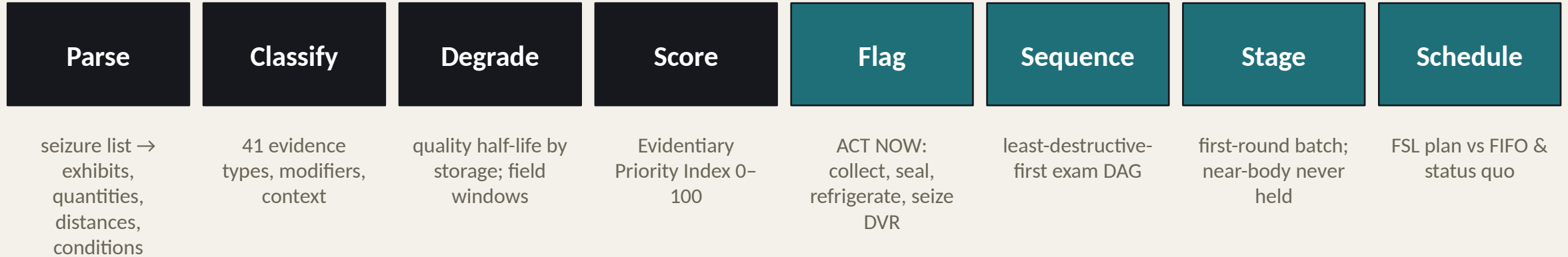
Examinations interfere

Powdering for prints can destroy touch DNA; destructive tests must come last.

It must hold up in court

Black-box scores are unusable. Every decision must be explainable, reproducible and logged — without personal data.

How Pramaan works — one deterministic engine



Outputs ranked queue with reasons • gap analysis (missing reference samples, CCTV canvass, photo coverage) • FSL submission packet (PDF) • SHA-256 hash-chained custody ledger

Surfaces IBM Bob via the MCP server (21 tools) • web dashboard + REST API • CLI — all sharing one live database. Optional watsonx.ai Granite may parse prose and pick a type from a closed vocabulary; it never scores.

Every priority is explainable, factor by factor

$$EPI = 100 \times (w_p \cdot P + w_u \cdot U + w_r \cdot R)$$

- P probative — crime-type relevance × individualizing potential × linkage (distance to body, recovered from accused)
- U urgency — hours until quality or the collection window is lost
- R irreplaceability — can it ever be re-collected?
- Weights per crime type; BNSS §184 samples floored at 85
- Each exhibit carries its caveat (what it does NOT prove) and what to corroborate it with

P PRAMAAN | evidence triage · FSL scheduling · chain of custody

+ New triage

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FSL network scale-out

Benchmark 200 exhibits

MCP server tools

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P1 3 · burglary · 21581
- Cr. No. 512/2026, PS Madha...
P1 5 · sexual assault · 828A4

HOMICIDE / SUSPICIOUS DEATH · PRM-20260928-B4ECB

Cr. No. 784/2026, PS Shamshabad Rural (synt)

Roadside homicide with burning — culvert under service road

EXHIBITS 18

P1 CRITICAL 8

P2 HIGH 8

HELD · STAGE 2 1

Act now — 6 field / preservation actions

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- EX-A12** **COLLECT_NOW** Still at the scene: evidence is likely lost in ~36 h (overwrite, weathering, washing). Collect immediately.
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Evidence queue 18

Lab plan 43

Gaps 4

Authenticity · BSA 3

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3	EX-A5	empty petrol bottle (1 litre) found 5 m from the body	Accelerant container (petrol / kerosene)	P1
4	EX-A17	viscera preserved at	Post-mortem viscera / stomach	P1

Degradation

Quality now 90% (risk threshold 60%)

Condition ambient · viscera profile

Usable for 42 h → 17 d if refrigerated

What if stored as ☐ refrigerated ☐ frozen ☐ dry_sealed ☐ ambient

Ex-A17 → refrigerated: usable window 42 h → 22 d; EPI 54.3 → 44.7; tier P1 → P3; case value retained 96.1% → 96.6%. (what-if only — nothing saved)

Examination sequence (least destructive first)

- Volatile analysis (ethanol, cyanide, phosphine — headspace GC) · TOXICOLOGY · 4 h **DESTRUCTIVE**
- Toxicology screening (drugs, pesticides, poisons) · TOXICOLOGY · 12 h **DESTRUCTIVE**

Flags

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VOLATILE_TOXICANTS Volatile compounds (alcohol, cyanide, phosphine, accelerants) are lost with delay — request headspace/volatile analysis first.

COLD_CHAIN_REQUIRED Maintain cold chain (2-8 °C) from collection to FSL receipt; record temperatures.

Interpretation guardrails

Does not prove A negative screen does not exclude poisoning (unscreened compounds, decomposition, delay).

Corroborate with Post-mortem findings; Suspected poison container; Treating hospital records

Handling Preserve in saturated salt solution (not for alcohol/volatile cases) in sealed glass jars; refrigerate; forward urgently.

Officer override (logged)

P1 Reason (required, written to ledger) **Apply**

Chain-of-custody event

sealed ☒ from (role) to (role)

Act now: degradation-aware urgency

P

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MCP /mcp

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P18 · homicide · B4ECB

Cr. No. 877/2026, PS Gachib...

P13 · hit and run · 29721

Cr. No. 1093/2026, PS Kukat...

P13 · burglary · 21561

Cr. No. 512/2026, PS Madha...

P15 · sexual assault · 82E44

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P1 CRITICAL8

P2 HIGH8

HELD · STAGE 21

URGENT FLAGS8

CUSTODY DEADLINE25 Dec 2026

EVIDENTIAL VALUE RETAINED · P1 DAYS96.1% vs 96.0% status quo · 2.06 vs 2.38 d

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FSL packet

AllP1P2P3Held

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- Viscera at room temperature: usable ~42 h — refrigerated ~17 days (quantified what-if)
- CCTV DVR not seized: overwrite risk → seize or export with hash today (BSA §63)
- Tyre and footwear impressions: rain shrinks the window six-fold
- Volatile accelerants: seal in an airtight can, not paper
- Examinations sequenced so touch-DNA swabbing precedes powdering

Staged testing + a scheduler that is measured, not claimed

Recommended plan

Recommended policy atc(k=0.5): evidential value retained 96.1% (perishable exhibits 94.6%) vs 96.0% (94.4%) examining everything in listing order and 96.0% for a static score sort; late exhibits 0 vs 0. P1 results in 2.06 working days on average vs 2.38. Stage-1 examinations complete in 11.0 working days; 1 exhibit is held for stage 2 (6 bench-hours not queued ahead of other cases).

PRAMAAN atc(k=0.5)

value retained **96.1%**
perishable value **94.6%**
late exhibits **0**
P1 mean (days) **2.06**
makespan (days) **11**

FIFO fifo

value retained **96.0%**
perishable value **94.4%**
late exhibits **0**
P1 mean (days) **2.38**
makespan (days) **10.8**

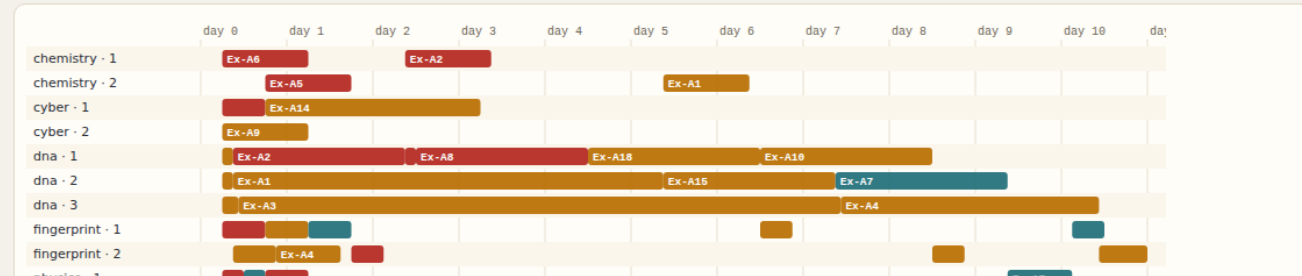
STATIC SORT priority

value retained **96.0%**
perishable value **94.5%**
late exhibits **0**
P1 mean (days) **2**
makespan (days) **10.6**

STATUS QUO status_quo

value retained **96.0%**
perishable value **94.4%**
late exhibits **0**
P1 mean (days) **2.38**
makespan (days) **10.8**

Examination timeline — Pramaan (click a policy card to compare; bars coloured by tier, faded = quality below 70% when examined)



- Many near-identical exhibits? The 3 most promising go first; the rest are held for stage 2
- Proximity trumps volume — traces within 10 m of the body are never held
- Job-shop simulation across divisions and examiners; Apparent-Tardiness-Cost portfolio
- Quality due dates and the BNSS §187(3) report deadline drive the order
- Always shown beside status quo, FIFO and static-sort baselines

RESULTS

Hyderabad-scale benchmark: 200 exhibits, 5 hidden needles

93.2%

vs 85.5%

evidential value retained

0

vs 13

exhibits analysed late

6.6 d

vs 29.3 d

P1 results (working days)

5 / 5

needles

surfaced as P1 (worst rank 36/200)

- Baseline = every exhibit examined in listing order (status quo). Staging removes 124 duplicates from the first queue; ordering speeds up P1 results.
- Needles: the half-burnt butt 3 m from the body, viscera at ambient, uncollected toll-plaza CCTV, fingernail scrapings, reference blood.
- Honest caveats: synthetic data and heuristic half-lives/capacities — reproducible with ``python -m pramaan benchmark``; 46 automated tests in CI.

Scales from one lab to the state's forensic network

2000

exhibits from 10 cases

triaged in 4.51 s (443/s), one laptop core

38.39 d

vs 84.85 d

P1 results: network plan vs everything to State FSL HQ

312.0 d

vs 492.0 d

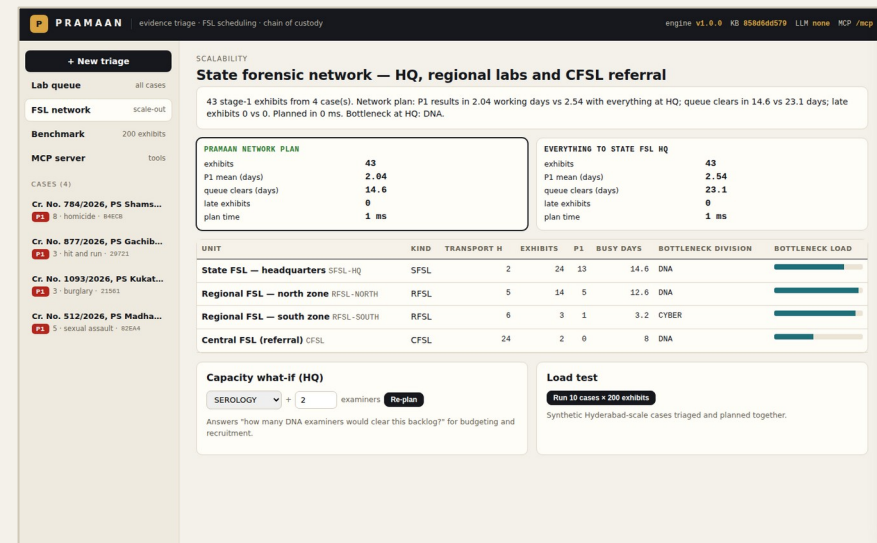
queue clears across 788 stage-1 exhibits

10 ms

planning time

heap timelines: $O(J \cdot U \cdot \text{ops} \cdot \log E)$

- State FSL HQ + Regional FSLs + CFSL referral (P1/P2 only) planned as ONE resource pool (laboratory.yaml → network:)
- Each exhibit goes to the unit where it completes earliest, including transport; all its exams stay on one object
- Capacity what-if per division (+N DNA examiners) for recruitment and budget requests; bottleneck shown per unit
- One queue across all open cases: a perishable exhibit from today overtakes a stable one from last week
- Stateless API, stdio or HTTP MCP, Docker, no CDN — runs in air-gapped district labs



IBM Bob is the officer's agent — the engine stays in charge

Officer

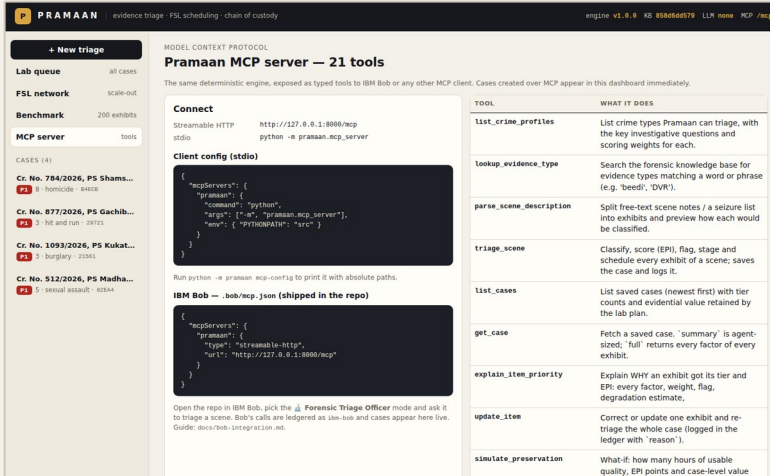
IBM Bob agent loop
AGENTS.md · skills · modes

Pramaan MCP
21 tools · /mcp

Engine + KB
+ ledger

state.json · audit.jsonl
report.html · graph.html

- AGENTS.md — the agent protocol (IBM Bob's counterpart of SAVVYDFIR's CLAUDE.md)
- 7 skills in .bob/skills/ — 7-phase evidence-triage-workflow, classification, preservation & custody, FSL scheduling, court-ready outputs, BSA authenticity, tools reference
- 4 custom modes —  Forensic Triage (edits only case-notes/),  FSL Liaison,  Court Reviewer,  Legal Cell (BSA); rules + slash commands
- 5 lifecycle hooks — briefing, privacy block, evidence-store write guard, MCP audit, Stop completion gate
- .bob/mcp.json → Pramaan MCP; Bob never invents a score



PRAMAAN evidence triage · FSL scheduling · chain of custody engine v1.0.0 KB 8586d6579 LLN none MCP /mcp

+ New triage

Lab queue all cases

FSL network scale-out

Benchmark 200 exhibits

MCP server tools

CASES (4)

- Cr. No. 784/2026, P5 Shams...
P5 0 - homicide - 1400A
- Cr. No. 877/2026, P5 Gachib...
P5 3 - hit and run - 29721
- Cr. No. 1093/2026, P5 Kukat...
P5 3 - burglary - 21061
- Cr. No. 512/2026, P5 Madha...
P5 5 - sexual assault - 42844

MODEL CONTEXT PROTOCOL

Pramaan MCP server — 21 tools

The same deterministic engine, exposed as typed tools to IBM Bob or any other MCP client. Cases created over MCP appear in this dashboard immediately.

TOOL	WHAT IT DOES
list_crime_profiles	List crime types Pramaan can triage, with the key investigative questions and scoring weights for each.
lookup_evidence_type	Search the forensic knowledge base for evidence types matching a word or phrase (e.g. 'beddi', 'DVR').
parse_scene_description	Split free-text scene notes / a seizure list into exhibits and preview how each would be classified.
triage_scene	Classify, score (EPI), flag, stage and schedule every exhibit of a scene; saves the case and logs it.
list_cases	List saved cases (newest first) with tier counts and evidential value retained by the lab plan.
get_case	Fetch a saved case. 'summary' is agent-sized; 'full' returns every factor of every exhibit.
explain_item_priority	Explain WHY an exhibit got its tier and EPI: every factor, weight, flag, degradation estimate.
update_item	Correct or update one exhibit and re-triage the whole case (logged in the ledger with 'reason').
simulate_preservation	What-if: how many hours of usable quality. EPI points and case-level value does moving one exhibit to

Connect

Streamable HTTP `http://127.0.0.1:8080/mcp`
stdio `python -m pramaan.mcp_server`

Client config (stdio)

```
{
  "mcpServers": {
    "pramaan": {
      "command": "python",
      "args": [
        "-m",
        "pramaan.mcp_server"
      ],
      "env": {
        "PYTHONPATH": "src"
      }
    }
  }
}
```

Run `python -m pramaan.mcp-config` to print it with absolute paths.

IBM Bob — .bob/mcp.json (shipped in the repo)

```
{
  "mcpServers": {
    "pramaan": {
      "type": "streamable-http",
      "url": "http://127.0.0.1:8080/mcp"
    }
  }
}
```

Open the repo in IBM Bob, pick the `g` Forensic Triage Officer mode and ask it to triage a scene. Bob's calls are ledgered as `lab-solve` and cases appear here live. Guide: docs/bob-integration.md

Law enforcement: is the evidence authentic, reliable and provable?

BSA s.61

Electronic records recognised; not refused merely for being electronic

BSA s.62

Contents proved under s.63 — route: original device vs computer output

BSA s.63(2)

(a) regular use · (b) ordinary course · (c) proper operation · (d) derived output

BSA s.63(4)

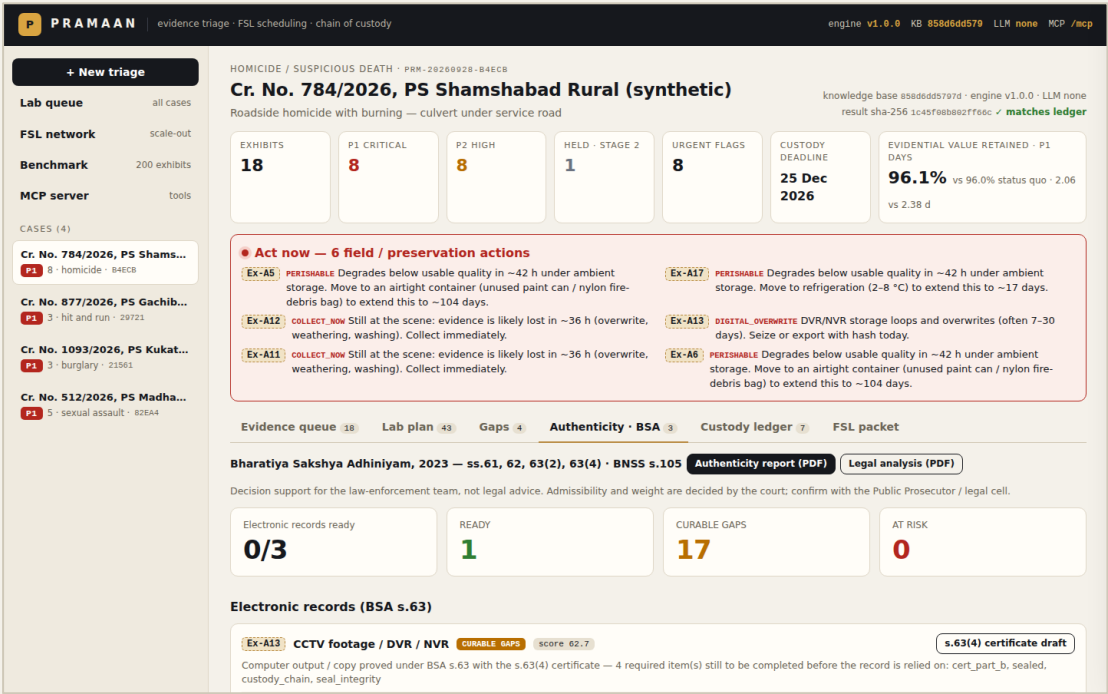
Schedule certificate: Part A (person in charge) + Part B (expert) with hash

Integrity

FSL re-hash must equal seizure hash — mismatch = AT RISK

BNSS s.105

Search & seizure recorded on video; custody seals at every hand-over



READY · CURABLE GAPS · AT RISK per exhibit, remedies by role (IO, device in-charge, FSL expert, legal cell), authenticity report PDF, pre-filled s.63(4) certificate drafts. Legal flags never change forensic priority; the court decides admissibility.

Integrity, privacy and the law — enforced in code

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Gaps 4

Authenticity · BSA 3

Custody ledger 7

FSL packet

Verify entire hash chain

Each entry's SHA-256 covers its content and the previous entry's hash — editing, deleting or re-ordering any row breaks the chain.

Chain intact — 10 entries, head b8b2852ac3583b97c03a5ba8...

SEQ	TIME (UTC)	ACTOR	ACTION	ITEM	DETAILS	HASH PREVIEW
10	2026-09-28 06:47:13	dashboard	custody:handed_over	Ex-002	{"from":"IO","label":"Ex-A2","note":"","seal_intact":true,"to":"Malkhana"}	b8b2852a-efabb496
9	2026-09-28 06:47:13	dashboard	custody:sealed	Ex-002	{"from":"IO","label":"Ex-A2","note":"","seal_intact":true,"to":null}	efabb496-27a8859c
8	2026-09-28	dashboard	update_reason	Ex-A13	{"changes":{"electronic":{"acquisition":"exported copy","certificate part a":true,"certificate part b":false,"	27a8859c-629be33f

- SHA-256 hash-chained ledger: edit one row and verification pinpoints it
- Knowledge-base hash + result hash on every run and every packet page
- Privacy gate on every surface: Aadhaar, phone, e-mail, hashed protected names (BNS §72, DPDP Act)
- Officer overrides need a reason and are logged
- Deterministic engine: same input + same KB = same ranking

Anchored in the code — evaluation criteria

Technical implementation quality

- `scoring.py` — explainable EPI, per-factor points
- `degradation.py` · `sequencing.py` (Kahn DAG)
- `scheduler.py` — FIFO/static/ATC portfolio with baselines
- `network.py` — heap timelines, capacity what-if
- `bsa.py` — BSA ss.61-63 checks, Schedule certificate PDF
- `store.py` — SHA-256 hash-chained ledger; `privacy.py` gate
- `mcp_server.py` — 21 typed tools; `api.py`; `cli.py`; `web/`
- 46 tests + CI; `.bob` contract tests

Innovation & differentiation

- Staged testing with a ≤ 10 m proximity exemption (Hyderabad butt stays in batch 1)
- Urgency = hours to evidential-quality loss; preservation what-ifs
- Custody-deadline (BNSS s.187(3)) aware scheduling
- Legal readiness decoupled from forensic priority (tested)
- Certificates drafted from structured facts + hash verification
- Network scale-out: earliest completion incl. transport; CFSL for P1/P2
- LLM proposes, rules dispose (proved by test)
- IBM Bob guard-rails as code: hooks + completion gate

WHAT'S NEXT

From hackathon to a State FSL pilot

Calibrate

Half-lives, exam hours and examiner counts with a partner FSL; the KB hash versions every change.

See photos

watsonx.ai vision to flag exhibits visible in scene photos but missing from the list.

Integrate

Seizure data from e-Sakshya / CCTNS; FSL LIMS hand-off; PostgreSQL + SSO for state-wide scale.

Anchor

RFC 3161 timestamps on the ledger head for court-grade custody proofs.

Run it: `docker compose up --build` → `http://127.0.0.1:8000` · Bob guide: `docs/bob-integration.md`

Team Pramaan · Pramaan means "proof"